



SIMATS ENGINEERING

SOLVED PROBLEMS

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COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

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1.Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.

Answer:

Given data

12, 15, 18, 15, 20, 22, 15, 25

Step 1: Arrange the data

Arrange the values in ascending order:

12, 15, 15, 15, 18, 20, 22, 25

Number of observations: $n = 8$

Mean

The formula for mean is:

$$\text{Mean} = \sum x / n$$

Sum of all values: $12+15+18+15+20+22+15+25=142$

Therefore, $\text{Mean} = 142 / 8 = 17.75$

Median

Since there are 8 observations, the median is the average of the 4th and 5th values.

4th value = 15

5th value = 18

$\text{Median} = 15 + 18 / 2 = 16.5$

Mode

The mode is the value that occurs most frequently.

- 15 occurs **3 times**
- Other values occur only once

Therefore: $\text{Mode} = 15$

Range

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$\text{Range} = 25 - 12 = 13$$

Final Answer

- **Mean = 17.75**
- **Median = 16.5**
- **Mode = 15**
- **Range = 13**

2. Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.

Answer:

Given data

2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4 **Step**

1: Identify the unique values

The values present are:

2, 3, 4, 5, 6

Step 2: Count how many times each value occurs

Value	Frequency
2	3
3	3
4	3
5	2
6	1
Total	12

Verification

$$3+3+3+2+1=12$$

The total frequency is equal to the total number of observations.

Final Answer

The most frequently occurring values are 2, 3 and 4, each with a frequency of 3.

3.Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.

Answer:

Given data

5, 8, 10, 12, 15, 18, 20, 22, 25, 30

The data is already arranged in ascending order.

Number of observations: $n=10$

Step 1: Find Q2

Q2 is the median.

Since there are 10 values:

5, 8, 10, 12, **15, 18**, 20, 22, 25, 30

$$Q2 = 15 + 18 / 2 = 16.5$$

Step 2: Find Q1

The lower half is:

5, 8, 10, 12, 15

The middle value is: $Q1 = 10$

Step 3: Find Q3

The upper half is:

18, 20, 22, 25, 30

The middle value is: $Q3 = 22$

Step 4: Calculate IQR

The Interquartile Range is: $IQR = Q3 - Q1$

$$IQR = 22 - 10 = 12$$

Final Answer

- **$Q1 = 10$**
- **$Q2 = 16.5$**
- **$Q3 = 22$**
- **$IQR = 12$**

4. Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.

Answer:

Given data

10, 12, 13, 15, 16, 17, 18, 20, 45

The data is already arranged.

Step 1: Find Q1 Lower

half:

10, 12, 13, 15

Since there are 4 values:

$$Q1 = 12 + 13 / 2$$

$$Q1 = 12.5$$

Step 2: Find Q3

Upper half:

16, 17, 18, 20

$$Q3 = 17 + 18 / 2$$

$$Q3 = 17.5$$

Step 3: Calculate IQR

$$IQR = Q3 - Q1$$

$$IQR = 17.5 - 12.5$$

$$IQR = 5$$

Step 4: Calculate Lower Fence Formula:

$$\text{Lower Fence} = Q1 - 1.5(IQR)$$

$$= 12.5 - 1.5(5)$$

$$= 12.5 - 7.5$$

$$= 5$$

Step 5: Calculate Upper Fence

$$\text{Upper Fence} = Q3 + 1.5(IQR)$$

$$= 17.5 + 1.5(5)$$

$$= 17.5 + 7.5$$

$$= 25$$

Step 6: Identify the outlier Any

value:

- Less than 5, or
- Greater than 25 is considered an outlier.

The given data contains 45.

$$45 > 25$$

Therefore:

Final Answer

45 is an outlier.

5. Determine the skewness (Asymmetry) of: 5, 6, 7, 7, 8, 8, 9, 20.

Answer:

Given data

5, 6, 7, 7, 8, 8, 9, 20

Step 1: Calculate the Mean

$$\text{Mean} = 5 + 6 + 7 + 7 + 8 + 8 + 9 + 20 / 8$$

$$\text{Mean} = 70 / 8$$

$$\text{Mean} = 8.75$$

Step 2: Calculate the Median There

are 8 observations, so:

$$\text{Median} = 4\text{th value} + 5\text{th value} / 2$$

$$4\text{th value} = 7$$

$$5\text{th value} = 8$$

$$\text{Median} = 7 + 8 / 2$$

$$\text{Median} = 7.5$$

Step 3: Compare Mean and Median

$$\text{Mean} = 8.75$$

$$\text{Median} = 7.5 \text{ Therefore:}$$

$$\text{Mean} > \text{Median}$$

When the mean is greater than the median, the distribution generally has a positive/right skew.

The value 20 is much larger than the other observations and pulls the distribution toward the right.

Pearson's Coefficient of Skewness

If required:

$$\text{Skewness} = 3(\text{Mean} - \text{Median}) / \text{Standard Deviation}$$

The standard deviation is approximately 4.41.

Therefore:

$$\text{Skewness} = 3(8.75 - 7.5) / 4.41$$

$$= 3.75 / 4.41$$

$$\approx 0.85$$

Final Answer

The dataset is positively skewed (right-skewed).

Reason: The extreme value 20 pulls the distribution toward the right

6.A uniformly distributed variable ranges from 10 to 50. Find $P(20 < X < 35)$.

Answer:

Given:

- Minimum = 10
- Maximum = 50
- Find $P(20 < X < 35)$

For a uniform distribution:

$$P(a < X < b) = (b-a)/(\text{Maximum}-\text{Minimum})$$

$$= (35-20)/(50-10)$$

$$= 15/40$$

$$= 0.375$$

Percentage:

$$0.375 \times 100 = 37.5\%$$

$$\text{Answer: } P(20 < X < 35) = 0.375 \text{ or } 37.5\%$$

7.A normal distribution has mean 100 and standard deviation 15. Find the Z-score for $X=130$.

Answer:

Given:

- Mean $\mu = 100$
- Standard deviation $\sigma = 15$
- $X = 130$ Formula:

$$Z = (X - \mu) / \sigma$$

$$= (130 - 100) / 15$$

$$= 30/15$$

$$= 2$$

$$Z\text{-score} = 2$$

This means 130 is 2 standard deviations above the mean.

8. For the same normal distribution, find $P(X < 130)$.

Answer:

From Question 7:

$$Z = 2$$

From the standard normal distribution table:

$$P(Z < 2) = 0.9772$$

Therefore: $P(X <$

$$130) = 0.9772$$

Percentage:

$$97.72\%$$

$$P(X < 130) = 0.9772 \text{ or } 97.72\%$$

9. Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.

Answer:

Consider the dataset:

10, 12, 13, 14, 15, 16, 18, 100

Here, 100 is an extreme value.

Before treatment

Mean:

$$= (10+12+13+14+15+16+18+100)/8$$

$$= 198/8 =$$

$$24.75$$

Median:

$$= (14+15)/2$$

$$= 14.5$$

After removing the outlier 100

Data:

10, 12, 13, 14, 15, 16, 18

Mean:

$$= 98/7$$

$$= 14$$

Median:

$$14$$

Measure	Before	After
Mean	24.75	14
Median	14.5	14

The outlier greatly increases the mean, while the median changes only slightly. Therefore, the median is more resistant to outliers.

10. Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.

Answer:

Consider the student marks:

45, 50, 52, 55, 60, 62, 65, 68, 70, 95

Step 1: Central Tendency

Mean

$$= 622/10$$

$$= 62.2$$

Median

$$= (60+62)/2$$

$$= 61$$

Mode

No value occurs more than once.

Mode = No mode

Step 2: Dispersion

Range

$$= \text{Maximum} - \text{Minimum}$$

$$= 95 - 45$$

$$= 50$$

Standard Deviation ≈ 13.36

This indicates that the marks have a moderate amount of variation around the mean.

Step 3: Quartiles and IQR

Lower half:

45, 50, 52, 55, 60

$$Q1 = 52$$

Upper half:

62, 65, 68, 70, 95

$$Q3 = 68$$

$$IQR = Q3 - Q1$$

$$= 68 - 52$$

$$= 16$$

Step 4: Outlier Detection

Upper fence:

$$Q3 + 1.5(IQR)$$

$$= 68 + 1.5(16)$$

$$= 68 + 24$$

$$= 92$$

Lower fence:

$$Q1 - 1.5(IQR)$$

$$= 52 - 24$$

$$= 28$$

Since:

$$95 > 92$$

95 is an outlier.

Step 5: Distribution and Skewness

The mean is:

62.2

The median is:

61

Since $\text{Mean} > \text{Median}$, the data is positively skewed (right-skewed).

The high mark 95 pulls the distribution toward the right.

Final EDA Summary

Measure	Result
Mean	62.2
Median	61
Mode	No mode
Range	50
Standard Deviation	13.36
Q1	52
Q3	68
IQR	16
Outlier	95
Skewness	Positive / Right-skewed

Conclusion: The student marks show moderate variation and are positively skewed due to the extreme value 95. The IQR method identifies 95 as an outlier.